



MGSC 416-001/002: Data Driven Models for Operation Analytics

Professor Rim Harris

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Fantasy Football:

A Data-Driven Approach to Predicting Player Performance and Optimizing Lineups

Group 14:

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I. Introduction

Fantasy sports have evolved into a multi-billion-dollar digital industry, with fantasy football emerging as one of the most widely adopted formats, particularly through platforms such as ESPN. In fantasy football, participants act as team managers by drafting and managing players from the National Football League (NFL), earning points based on the real-world performance of those players. Each week, users must decide which players to start across various roles such as receiver, quarterback, running back, and tight end while considering factors like player form, opponent strength, and injury status. This dynamic, data-rich environment requires continuous decision-making under uncertainty, making it highly relevant for analytical modeling.

Despite the abundance of available data, many fantasy football decisions remain intuitive or heuristic-based rather than systematically optimized. Players often rely on expert opinions, recent performance trends, or personal biases, which may not lead to optimal outcomes. This creates an opportunity to apply business analytics techniques to improve decision quality. Specifically, the problem addressed in this project is how to leverage historical player performance data and contextual variables to both predict future player outputs and optimize weekly lineup decisions through trades. By doing so, the project aims to bridge the gap between data availability and effective decision-making in a competitive setting. This is achieved by projecting future player scoring using advanced statistics as predictors, then comparing these projections against consensus trade values to identify players the general public is overvaluing or undervaluing.

II. Data Overview

The data used in this project consists of five different datasets: one for each of the four fantasy football position groups (QBs, WRs, RBs, and TEs), and one more for consensus player trade value pre-week 11. The former are sourced from FantasyLife, a paid fantasy football data platform that collects advanced statistics not available to the general public. The WRs dataset has 2412 rows, RBs has 1563, TEs has 1459, and QBs has 631. Each one's columns consists of a mix of the following metrics:

- **Week:** The NFL week number for the game (ranging from 1-17). (QB, WR, RB, TE)
- **Players:** Player name. (QB, WR, RB, TE)
- **Snaps:** Total % of offensive snaps played. (QB, RB)
- **Pass ATT:** Pass attempts thrown by the quarterback. (QB)
- **Dropbacks:** Total dropbacks (pass attempts + sacks + scrambles). (QB)
- **ADOT:** Average Depth of Target. The average distance downfield of throws (QB) or of targets received (WR/TE). (QB, WR, TE)
- **COMP %:** Completion % on pass attempts. (QB)
- **YPA:** Yards Per Attempt on passes. (QB)
- **DES Rush ATT:** Designed rush attempts (planned, not scrambles) as a % of snaps. (QB)

- **Scrambles:** % of snaps where QB ran the ball after a designed pass play failed. (QB)
- **Sacks:** % of snaps where the QB was sacked. (QB)
- **I5 Rush ATT:** Player's share of rush attempts inside the opponent's 5-yard line (QB, RB)
- **EZ Rush ATT:** Player's share of all rush attempts in the end zone / red zone. (RB)
- **GL Rush ATT:** Player's share of all rush attempts at the goal line. (RB)
- **Rush ATT:** Total rushing attempts. (RB)
- **Routes:** % of pass routes run. (WR, RB, TE)
- **Targets:** Target share as a % of all QB passes. (WR, RB, TE)
- **TPRR:** Target share as a % of QB passes when the player was on the field (WR, RB, TE)
- **Catchable TGTS:** % of targets that were accurate enough to be caught. (WR, RB, TE)
- **Air Yards:** Total distance of all targets through the air before the catch point. (WR, TE)
- **EZ TGTS:** Player's share of all targets received in the end zone. (WR, TE)
- **3/4 Down TGTS:** Player's share of targets on 3rd/4th down (WR, TE)
- **PA TGTS:** % of player's targets on play-action pass plays. (WR, TE)
- **PPR:** Points Per Reception fantasy score for that week. (QB, WR, RB, TE)
- **PPR Rank:** Positional rank in PPR scoring for that week. (QB, WR, RB, TE)

The consensus player trade value dataset is sourced from CBS sports and represents their consensus view on each player's trade value pre-week 11. It consists of 118 rows, with columns being player name, the team they play for, and their trade value, ranging from 5-46. Significant data wrangling was done before the analysis. A participation filter then removed low-snap or low-route weeks to prevent inaccurate data points from injuries and garbage time (for example, QBs needed at least 75% of snaps). Since the desired model produces a single projection per player, the input data needed to match that structure. As a result, the remaining weeks 1-10 data was collapsed from weekly rows into one row per player by computing four summary features per statistic: mean, trend (slope of linear line of best fit), standard deviation, and last-3-week average.

III. Methodology / Model

Regression Analysis

The ultimate goal for our regression analysis was to build a robust predictive model that would train on data from the first 10 weeks to then predict player value and aggregate points from week 11 to 17. We first engineered per-player features such as mean, trend/slope, standard deviation and 3-week averages for each of the statistics relevant to each position. We then set each player's **Average PPR** as a target for the model to predict.

In our approach, we used lasso regression for Tight Ends, Running Backs and Wide Receivers, and ridge regression for Quarterbacks. Lasso was chosen because it selects only the most important player statistics and discards the rest, which works well when there are enough players to learn from. For Quarterbacks however, the limited sample size caused projections to be compressed to the mean, which pivoting to the Ridge model partially resolved.

We then performed cross-validation to evaluate model performance. We opted for both “Leave-One-Out” (LOOCV) and K-Fold CV, in our case 5 folds. Due to the small sample size of players per position which causes in of itself a lot of noise surrounding the estimate of model error, LOOCV was performed as a primary check. LOOCV ensures that the cross-validation always uses the largest possible training set, “n-1”. The 5-Fold CV was performed afterwards as a sanity check, to ensure that the high variance nature of the LOOCV was balanced out.

Optimization Model Analysis

In our approach, we combined our projected player performance with an optimization model in order to identify trades that improve the roster composition and most importantly the starting lineup strength. We start by generating a set of possible trade packages. Let R denote our current (starting) roster and F the set of available players. Each trade package t consists of players sent $S_t \subseteq R$ and players received $G_t \subseteq F$. We allow realistic trade formats of the form: 1 for 1, 2 for 2, 1 for 2 and 2 for 2. It is possible to turn on or off any of these trade formats before computing. For each package, total outgoing and incoming trade value and projected PPR are calculated. **The full mathematical formulation of trade logic can be found in figure A5.**

Then, we set up a binary optimization model that selects which trade packages to execute, finds the final 13-player roster, and chooses the optimal starting lineup. The model uses three decision variables: x_t indicating whether trade t is selected, y_i indicating whether player i is on the final roster, and z_i indicating whether player i is in the starting lineup. Our objective is to maximize the projected performance of the team, with most of the weight placed on starting lineup PPR and a smaller weight on bench value:

$$\max \sum_{i \in I} p_i z_i + 0.2 \sum_{i \in I} p_i (y_i - z_i)$$

Where:

- I is the set of all players,
- p_i is the projected PPR of player i ,
- $y_i \in \{0,1\}$ indicates if player i is on the final roster and
- $z_i \in \{0,1\}$ indicates if player i is selected in the starting lineup

The weight of bench players was decided through testing, ensuring that the model’s trades would not improve the starting lineup at great expense to the quality of the bench. Transitions from in and out of the roster are modeled explicitly so that final roster membership accounts for initial ownership, acquisitions, and departures from selected trades. The starting lineup is not fixed in advance; instead, it is

chosen by optimizing points from all of the team's players subject to positional requirements and total number of starters which are standard in Fantasy Football.

To ensure the model was realistic, we added constraints limiting the number of trades to a maximum of 5, limiting consensus value in to at least 80% of value out to eliminate unfair outlier trades, roster size and positional balance requirements, and restrictions preventing a player from being traded multiple times. Most importantly, value in was constrained to be less than or equal to value out in any given trade. In addition, because of poor Quarterback PPR projections, we added a constraint forbidding them from being traded. This model evaluates trades based on the impact they have on the best achievable starting lineup rather than focusing on player to player comparisons. This allows for more realistic and strategically meaningful recommendations. **The full mathematical formulation can be found in figure A6.**

IV. Results and Insights

Regression Analysis

We applied Lasso regression (TE, RB, WR) alongside Ridge regression (QB) to forecast each player's mean PPR from Weeks 11–17, derived from stats spanning Weeks 1–10. Every raw metric gets split into four different attributes: mean, trend (slope), standard deviation, and a rolling three-week average, alongside prior PPR, which dominated as the top predictor universally. The L1 penalty in Lasso eliminated insignificant variables, keeping only the most influential predictors (5 for TE, 10 for RB, 26 for WR). Ridge was preferred for Quarterbacks to prevent excessive shrinkage due to limited data ($n = 57$). Optimal regularization strength was auto-adjusted via LassoCV and RidgeCV. Running backs produced the strongest in-sample fit ($R^2 = 0.742$). Tight ends and wide receivers trailed closely ($R^2 = 0.649, 0.650$), while quarterbacks yielded the weakest results ($R^2 = 0.491$). Leave-One-Out CV validated the model's stability, with LOO MAE spanning 1.97 pts for Tight Ends to 4.99 pts for Quarterbacks. Tested against 118 players' actual Weeks 11–17 output, the overall MAE settled at 3.35 pts (RMSE = 4.29, $r = 0.636$), exhibiting that Tight Ends excelled in precision ($r = 0.667$), while Quarterbacks' performance was far from optimal ($r = 0.079$). Complete training and out-of-sample **results appear in Tables A1 and A2.**

Trade Value Gap and Player Mispricing

In addition to using Week 10 FantasyPros consensus trade values for comparison purposes to identify mispricings based on our models' projection, we can utilize this data to provide actionable misvalue comparisons to be used during future weeks (see **Table A3**). The strongest buy-lows were Wan'dale Robinson (Value: 6, Proj: 14.15, actual: 15.08), and Caleb Williams (Value: 7, Proj: 14.12, actual: 16.77), both underpriced relative to their usage trends. Conversely, our top four sell high recommendations were Jonathan Taylor (Value: 46, actual: 13.75), Drake London (Value: 34, actual:

8.67), and Emeka Egbuka (Value: 30, actual: 6.76); each of these players performed below their respective market valuations as expected. Our model's key miss was in its inability to accurately assess the Value of Jahmyr Gibbs (Value: 41, actual: 22.94). Josh Allen (Proj: 15.78, Actual: 22.99) clearly illustrates the limitations associated with projecting Quarterback ceilings.

Optimization Model

To demonstrate the full pipeline, the optimizer was applied to one of our own fantasy rosters: Drake Maye (QB); Josh Jacobs, Chase Brown, Kyle Monangai, Tyler Allgeier, and Jordan Mason (RB); Amon-Ra St. Brown, Stefon Diggs, Quentin Johnston, Ricky Pearsall, Alec Pierce, and DJ Moore (WR); and Jake Ferguson (TE). Gurobi optimized the solutions and identified the optimal set of five Value-balanced trades. Prior to trading, our model started from a 13-player initial roster with a pre-trade projected starting lineup of 93.93 PPR points. After trading, our model increased the projected lineup PPR to 124.30 pts (+30.37pts, +32%) greater than the original projected lineup. Importantly, our optimization model was able to successfully validate its results against actual week 11-17 performances. As shown above, the actual week 11-17 starting lineup PPR after trading was 129.25 while the actual week 11-17 starting lineup PPR prior to trading was 100.5. Therefore, the model resulted in a real-world increase of 28.75 PPR points (+28.6%). When stress tested on 100 different teams with mean trade value of 205 and standard deviation of 25, the model's trades resulted in an average uplift of 32.14 points on actual week 11-17 data, and improved the team's actual outcome in 99 out of 100 instances (see **figure A7**).

Conclusion and Limitations

This project demonstrates how data-driven modeling can significantly improve decision-making in fantasy football by replacing intuition-based strategies with systematic, and optimized approaches. Using Lasso and Ridge regression, we generated player performance projections based on historical usage and scoring trends, and embedded these into an optimization model that evaluates trades and constructs the optimal starting lineup. The results show that this integrated approach produces meaningful gains: projected starting lineup performance increased by over 30%, with a validated real-world improvement of 32.14 PPR points. From our results, four insights emerge:

First, a player's own recent PPR history dominates as the strongest predictor, confirming scoring momentum matters more than any individual usage metric. From a managerial perspective, this suggests that decision-makers should prioritize recent performance trends over isolated advanced metrics when making lineup or trade decisions. Second, the market reliably underprices low-value players with consistent usage trends, creating identifiable buy-low opportunities. This highlights the presence of systematic market inefficiencies that managers can exploit through disciplined, data-driven buy-low/sell-high strategies rather than relying solely on consensus rankings. Third, through multi-trade analysis, our optimization model has demonstrated that it is possible to make multiple trade moves

simultaneously that may result in improving team performance more significantly than making one trade move at a time. This reinforces the importance of evaluating decisions at a portfolio level, where jointly optimizing multiple moves lead to stronger overall roster formation. Fourth, and most importantly, the 32.14-point actual improvement in starting lineup PPR validates that the full pipeline process (project, compare, optimize) produces real-world gains and not just projected ones. Broadly, this demonstrates the importance of integrating predictive and prescriptive analytics to not only forecast outcomes but also improve division execution in practice.

There are four main limitations of the study that should be taken into account. First, the model does not directly account for changes related to recent injuries and other news affecting NFL team's depth charts, reducing its practical applicability in live settings. It also assigns an injured player the same value as a non-injured one, even if the injured player will play in fewer weeks. Second, players who suffered injuries earlier in the season did not have as much training data collected during weeks one through ten, leading to higher projection errors due to reduced sample sizes. The model also relies on historical usage patterns, which may not remain stable as player roles and team strategies evolve. At the same time, it only relies on the current season's data, and does not take into account a player's performance in previous seasons. Third, the model significantly undervalued elite quarterbacks' performance, likely because quarterback performance is heavily influenced by contextual factors such as opponent matchup and game flow that are not captured in the model.

An improved version of the model would project fantasy points in every week from 11-17 instead of a singular, aggregated projection. This would allow it to differentiate projections based on the strength of opponent matchups for different individual players. In addition, allowing users to sync their leagues, automatically importing their and all of their leaguemates' teams would simplify the model since it would be required to trade for players on the same team in 1-for-2 or 2-for-2 packages, decreasing run time and albeit likely slightly decreasing uplift potential.

Appendix

Table A1. Lasso/Ridge Training Performance (Features: Weeks 1–10 | Target: Avg. PPR Weeks 11–17)

Position	Model	N Players	Train R ²	Train MAE	Train RMSE	LOO CV MAE	Key Predictors
TE	Lasso	101	0.649	1.81	2.44	1.97	PPR_last3avg, Routes_mean, PPR_mean, Air Yards_mean, PPR_std
RB	Lasso	120	0.742	2.44	3.15	2.80	PPR_last3avg, Routes_last3avg, Rush ATT_mean, PPR_slope, TPRR_last3avg
WR	Lasso	182	0.650	2.35	3.06	2.87	Routes_mean, Routes_last3avg, PPR_mean, 3/4 Down TGTS_mean, Targets_slope
QB	Ridge	57	0.491	4.27	5.46	4.99	PPR_mean, Pass ATT_last3avg, Pass ATT_mean, Sacks_slope, PPR_last3avg

Table A2. Out-of-Sample Prediction Accuracy vs. Actual Weeks 11–17 Performance

Position	N	Test MAE (pts)	Test RMSE (pts)	r	Assesement
TE	17	2.93	3.97	0.677	Strong accuracy; stable usage patterns
RB	41	3.59	4.58	0.648	Best correlation among skill positions
WR	45	3.02	3.80	0.609	Largest sample; high week-to-week variance
QB	15	4.20	5.11	0.079	Elite QB ceilings systematically underestimated
Overall	118	3.35	4.29	0.636	All players with Week 10 trade value + actual PPR

Table A3. Trade Value Gap: Selected Player Mispricings (Week 10 Consensus Value vs. Model Projection)

Player	Position	Trade Value	Projected PPR	Actual PPR	Assessment	Verdict
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Wan'Dale Robinson	WR	6	14.15	15.08	Very low cost; model and actual aligned	Buy-Low
Caleb Williams	QB	7	14.12	16.77	Low market value; model and actual both strong	Buy-Low
Jonathan Taylor	RB	46	19.89	13.75	Highest value in dataset; model correctly skeptical	Sell-High
Drake London	WR	34	13.77	8.67	High market value; model skeptical — correctly so	Sell-High
Emeka Egbuka	WR	30	10.94	6.76	High value; model correctly projected modest output	Sell-High
Jahmyr Gibbs	RB	41	16.74	22.94	Model underestimated — missed second-half surge	Mixed
Brock Bowers	TE	21	12.34	14.77	Model and market well-aligned	Fair Value

Josh Allen	QB	25	15.78	22.99	Elite QB ceiling underestimated by model	Model limit
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Table A4. Optimizer-Selected Trade Packages and Actual Performance Validation

Send	Receive	Value Out	Value In	Projected Gain	Actual Gain	Assesment
Josh Jacobs + DJ Moore	Chris Olave + Trey McBride	40	38	+8.27	+17.52	Near even value
Kyle Monangai + Jordan Mason	Wan'Dale Robinson + Juwan Johnson	12	12	+8.21	+10.08	Even value
Tyler Allgeier + Jake Ferguson	Bucky Irving + Terry McLaurin	16	16	+9.02	+8.91	Even value
Amon-Ra St. Brown + Ricky Pearsall	De'Von Achane + Isiah Pacheco	48	48	+6.26	-0.32	Even value
Quentin Johnston + Alec Pierce	Omarion Hampton + Parker Washington	18	18	+5.24	+7.80	Even value
NET IMPROVEMENT	93.93 → 124.30 (proj) 100.50 → 129.25 (actual)	/	/	+30.37	+28.75	+32% / +29%

Figure A5. Trade Package Logic

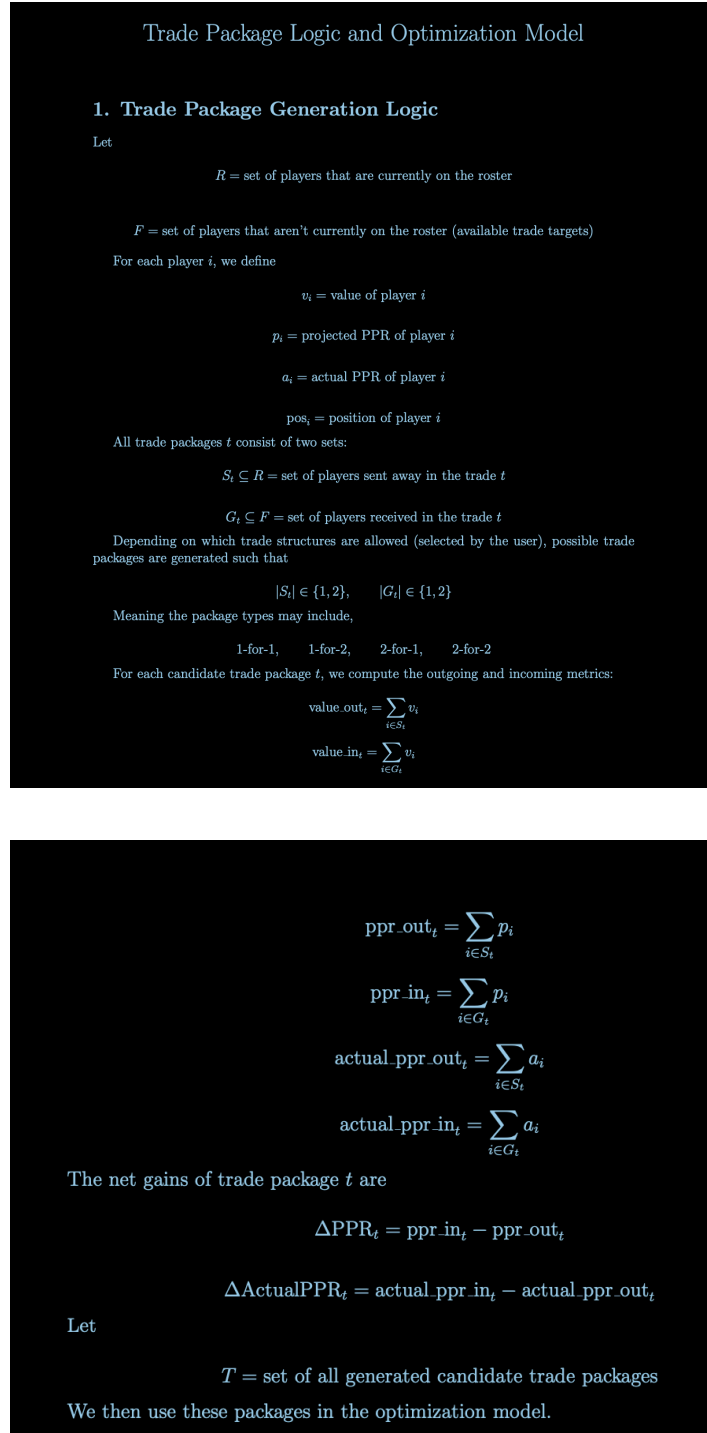


Figure A6. Optimization Model Formulation

2. Optimization Model

2.1 Sets and Indices

Let

I = set of all players in our dataset

T = set of all candidate trade packages

$R \subseteq I$ = set of players that are initially on the roster

$F \subseteq I$ = set of players that are initially not on the roster

For each trade package $t \in T$,

$S_t \subseteq R$ = players sent away in trade t

$G_t \subseteq F$ = players received in trade t

Defining the initial roster indicator:

$$r_i = \begin{cases} 1 & \text{if player } i \text{ is initially on my roster} \\ 0 & \text{otherwise} \end{cases}$$

Let

K = maximum number of trades allowed

2.2 Parameters

For each player $i \in I$,

v_i = trade value of player i

p_i = projected PPR of player i

$\text{pos}_i \in \{\text{QB, RB, WR, TE}\}$

For each trade package $t \in T$,

$$V_t^{\text{out}} = \sum_{i \in S_t} v_i$$

$$V_t^{\text{in}} = \sum_{i \in G_t} v_i$$

2.3 Decision Variables

For each trade package $t \in T$,

$$x_t = \begin{cases} 1 & \text{if trade package } t \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

For each player $i \in I$,

$$y_i = \begin{cases} 1 & \text{if player } i \text{ is on the final roster} \\ 0 & \text{otherwise} \end{cases}$$

$$z_i = \begin{cases} 1 & \text{if player } i \text{ is in the final starting lineup} \\ 0 & \text{otherwise} \end{cases}$$

2.4 Objective Function

The model has for objective to maximize the projected PPR of the final starting lineup, with a small weighted addition coming from bench players:

$$\max \sum_{i \in I} p_i z_i + 0.3 \sum_{i \in I} p_i (y_i - z_i)$$

The first term ensures that final starting lineup is rewarded. The second term gives partial value to players who remain on the bench.

2.5 Constraints

Trade limit

$$\sum_{t \in T} x_t \leq K$$

Roster players can be traded away once

$$\sum_{t \in T: i \in S_t} x_t \leq 1 \quad \forall i \in R$$

Non-roster players can be acquired once

$$\sum_{t \in T: i \in G_t} x_t \leq 1 \quad \forall i \in F$$

No quarterback trades allowed All trade packages containing QB's are forbidden since the projection for QB's were poor.

$$x_t = 0 \quad \forall t \in T \text{ such that there exists } i \in S_t \cup G_t \text{ with } \text{pos}_i = \text{QB}$$

Trade fairness constraints Incoming trade value has to be inferior or equal to outgoing trade value or it would not be feasible:

$$V_t^{\text{out}} x_t \geq V_t^{\text{in}} x_t \quad \forall t \in T$$

Incoming trade value must be at least 80% of outgoing trade value or the trade would seem very suspicious, and this also account for outlier projections:

$$V_t^{\text{in}} + M(1 - x_t) \geq 0.8 V_t^{\text{out}} \quad \forall t \in T$$

where M is a large enough constant.

Final roster membership For every player $i \in I$, the final roster indicator is equal to initial roster indicator plus acquisitions minus departures:

$$y_i = r_i + \sum_{t \in T: i \in G_t} x_t - \sum_{t \in T: i \in S_t} x_t \quad \forall i \in I$$

Final roster size

$$\sum_{i \in I} y_i = 13$$

Final roster composition

$$\sum_{i \in I: \text{pos}_i = \text{WR}} y_i \geq 4$$

$$\sum_{i \in I: \text{pos}_i = \text{RB}} y_i \geq 3$$

$$\sum_{i \in I: \text{pos}_i = \text{QB}} y_i \leq 2$$

$$\sum_{i \in I: \text{pos}_i = \text{TE}} y_i \leq 2$$

Starters must be on the final roster

$$z_i \leq y_i \quad \forall i \in I$$

Starting lineup size

$$\sum_{i \in I} z_i = 7$$

Starting lineup composition Exactly one quarterback:

$$\sum_{i \in I: \text{pos}_i = \text{QB}} z_i = 1$$

At least two running backs:

$$\sum_{i \in I: \text{pos}_i = \text{RB}} z_i \geq 2$$

At least two wide receivers:

$$\sum_{i \in I: \text{pos}_i = \text{WR}} z_i \geq 2$$

At least one tight end:

$$\sum_{i \in I: \text{pos}_i = \text{TE}} z_i \geq 1$$

FLEX structure The remaining six starters must come from the set of RB, WR, and TE players:

$$\sum_{i \in I: \text{pos}_i \in \{\text{RB}, \text{WR}, \text{TE}\}} z_i = 6$$

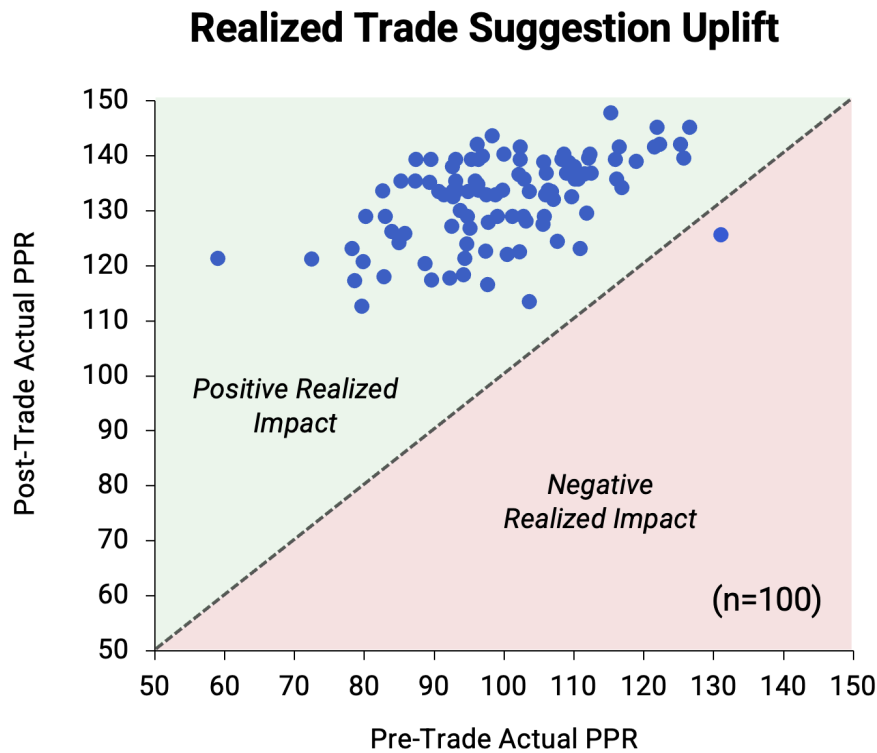
2.6 Variable Domains

$$x_t \in \{0, 1\} \quad \forall t \in T$$

$$y_i \in \{0, 1\} \quad \forall i \in I$$

$$z_i \in \{0, 1\} \quad \forall i \in I$$

Figure A7. Optimization Model Results on 100 Team Sample



References:

Fantasy sports business analysis report 2024–2030. (2024, October 30). *Yahoo Finance*.

<https://finance.yahoo.com/news/fantasy-sports-business-analysis-report-154500108.html>

Raw Datasets:

Receivers Dataset:

https://mcgill-my.sharepoint.com/:x:/g/personal/jeremie_nathaniel_mail_mcgill_ca/IQCn4s4nBAqeRZu3uejVlXIjAYHCRJi2g1C8FJ4tZGqEssI?e=Pjnwpc

Quarterbacks Dataset:

https://mcgill-my.sharepoint.com/:x:/g/personal/jeremie_nathaniel_mail_mcgill_ca/IQD0wn2LqkJbSpbZRDMfuQqaAYAxgPEK6l0rmUdmgG6QnB0?e=IBhztM

Runningbacks Dataset:

https://mcgill-my.sharepoint.com/:x:/g/personal/jeremie_nathaniel_mail_mcgill_ca/IQActcbkIbbLSbaOIJO2nlPaAQRyI4LkMMzxynJSUvtNH-M?e=ORixfN

Tight Ends Dataset:

https://mcgill-my.sharepoint.com/:x:/g/personal/jeremie_nathaniel_mail_mcgill_ca/IQCgqjEoxca3T5oeUrz-6Wm4Ac3sQJ1IPXk6kK2FvkPsahw?e=mf0NjKz

Pre-Week 11 Trade Value Chart:

https://mcgill-my.sharepoint.com/:x:/g/personal/jeremie_nathaniel_mail_mcgill_ca/IQC_3EZ3jfRxQLHQKLOKtoopAXDsoLOSEbk4QboOusnfbil?e=gb7SdS